

Early Sepsis Prediction from the First Six ICU Hours: Supervised and Unsupervised Exploration

Scott Pryde · project_sepsis · 10 October 2026 · Redevelopment period: five weeks

1 Proposal

The feasibility test confirmed that the PhysioNet/CinC Challenge 2019 dataset can support a leakage-safe pipeline that compares supervised and unsupervised AI methods on the same patients. This proposal redevelops that prototype over five weeks into a reproducible study and a basic application. The study predicts, from hours 0–5 of an ICU stay, whether a patient meets Sepsis-3 criteria within a fixed horizon, and explores label-free structure in the same data.

2 Rationale

Why this dataset. It is open access, large (40,336 ICU patients), labelled hourly against Sepsis-3, and drawn from two hospital systems, allowing external validation without new data agreements. Its informative missingness is realistic, and published Challenge results give a benchmark.

Why both method families. Supervised models answer “who will develop sepsis?”. Unsupervised methods ask whether that risk is already visible as structure, such as subgroups or unusual physiology, before any label is used.

What the feasibility test established.

- Data loads, validates and converts to one row per patient, with train-only preprocessing and patient-level splits.
- Four supervised models and a self-contained HTML report already run end to end.
- Issues to fix in redevelopment: an open-ended outcome window, inconsistent class weighting, no validation split, no cross-site test, and no unsupervised methods (tracked in `IMPROVEMENTS.md`).

3 Data source and sampling

ELEMENT	DEFINITION
Source	PhysioNet/CinC Challenge 2019, version 1.0.0, training sets A and B: 40,336 patients, 2,932 septic (7.3%), 40 hourly variables plus <code>SepsisLabel</code> . Records have at least 8 hours and are capped at two weeks.
Unit and timing	One prediction per patient at the end of hour 5. Inputs are hours 0–5 only.
Outcome	<code>SepsisLabel</code> = 1 within a fixed horizon after hour 5 (primary 24 h; sensitivity analyses at 12 h and 48 h). The label already switches on 6 h before onset. Negatives need complete follow-up for the horizon.
Splits	Stratified, patient-level, seeded 60 / 20 / 20 train / validation / test. External checks: train on A and test on B, then the reverse. Five repeated seeds for rank stability.
Uncertainty	1,000 paired bootstrap resamples of the test patients for every reported metric.
Development sample	A stratified 10% subsample (about 4,000 patients) for fast iteration. All reported results use the full data.

4 Synthetic data definition

A seeded generator (`synthetic.py`) produces files with the real schema, so tests, CI and the application can run without the real data. Synthetic results are never reported as findings.

PROPERTY	SPECIFICATION
Schema	Identical PSV columns and order (40 variables + <code>SepsisLabel</code>). Files are prefixed <code>s</code> so they cannot be mixed with real <code>p</code> files.
Population	Configurable size (default 2,000; 40,000 for scale tests), split 50/50 between two simulated sites with different baseline means and lab-ordering rates.
Stay length	Log-normal, bounded to 8–336 hours to match the real data's limits.
Risk and onset	Latent risk per patient sets the outcome via a logistic link, targeting 7% prevalence. Onset is at hour 4 or later; labels switch on 6 h before onset, as in PhysioNet.
Signals	Vitals drift with latent risks towards onset, plus autocorrelated noise. Three planted subgroups (hypotensive, febrile, respiratory) test whether clustering recovers known structure.
Missingness	Vitals 10–30% missing per hour; labs 80–98%, with ordering more frequent at higher risk (informative missingness).

5 Architecture and data pipeline

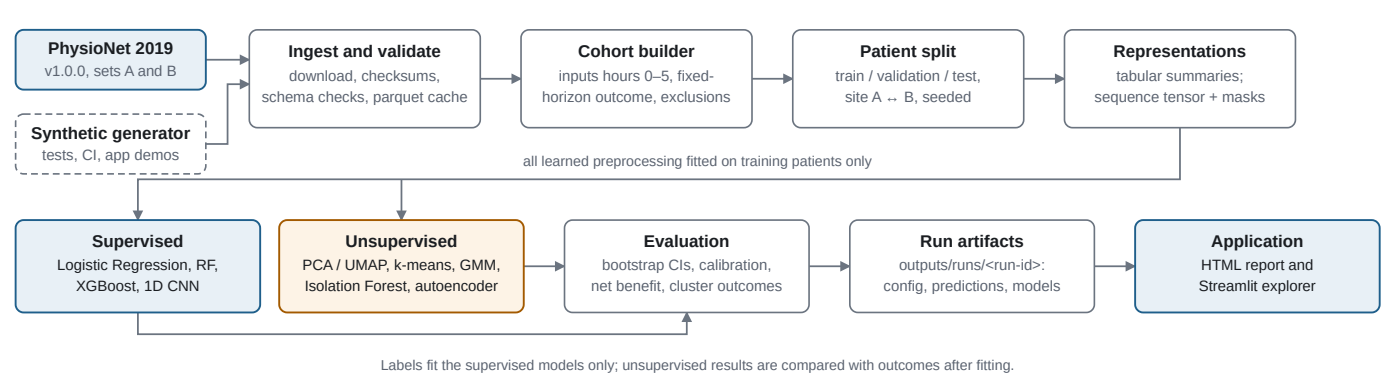


Figure 1. Pipeline from source data to application. Both method families read the same split and features and are scored by one evaluation module.

6 Environment and folder structure

A single Python 3.11 virtual environment holds both the ML pipeline and the application. Dependencies are split into optional extras so each part installs only what it needs.

```
py -3.11 -m venv .venv
.\venv\Scripts\Activate.ps1          # Windows
python -m pip install -U pip
pip install -e ".[ml,explore,app,dev]"
pip install -e ".[cnn]"               # PyTorch, optional
python scripts\download_data.py --flat
python scripts\make_synthetic.py --n 2000
pytest
sepsis-pipeline run --config configs\default.yaml
streamlit run app\Home.py
```

Extras: ml core pipeline; cnn PyTorch; explore UMAP; app Streamlit; dev tests and linting. .venv/, data/ and outputs/ stay out of Git.

Basic application. A read-only Streamlit app over saved run artifacts: cohort overview, model comparison, cluster explorer, and a patient viewer for synthetic patients. It never trains models.

```
project_sepsis/
├─ pyproject.toml  README.md  IMPROVEMENTS.md
├─ configs/       default.yaml, horizons.yaml
├─ data/          (git-ignored)
│  └─ raw/training_setA/ ... training_setB/
│  └─ processed/   parquet cache
│  └─ synthetic/
├─ src/sepsis_prediction/
│  └─ data.py cohort.py features.py splits.py
│  └─ synthetic.py
│  └─ supervised/   tabular.py cnn.py
│  └─ unsupervised/ embedding.py clustering.py
│  └─               anomaly.py autoencoder.py
│  └─ evaluation.py reporting.py cli.py
├─ app/
│  └─ Home.py
│  └─ pages/ 1_Cohort.py 2_Models.py
│             3_Clusters.py 4_Patient.py
├─ scripts/      download_data.py make_synthetic.py
├─ tests/        unit, pipeline, app smoke tests
├─ docs/         design, data card, model cards
└─ outputs/runs/<run-id>/ (git-ignored)
```

7 Models and libraries

FAMILY	MODELS	PURPOSE	LIBRARIES
Supervised, tabular	Logistic Regression, Random Forest, XGBoost	Interpretable baseline, non-linear ensemble, gradient boosting	scikit-learn, xgboost
Supervised, sequence	1D CNN with missingness masks	Tests whether hour-by-hour order adds signal	PyTorch
Unsupervised, structure	PCA, UMAP, k-means, Gaussian mixture	Label-free views and patient subgroups	scikit-learn, umap-learn
Unsupervised, anomaly	Isolation Forest, sequence autoencoder	Does “unusual” predict sepsis without labels?	scikit-learn, PyTorch
Data and evaluation	—	Loading, caching, statistics, figures, configs	pandas, NumPy, pyarrow, SciPy, matplotlib, PyYAML
Application and quality	—	Explorer app, tests, linting	Streamlit, pytest, ruff

Every model reports AUROC and AUPRC with bootstrap CIs, alert burden at a validation-chosen 80% sensitivity, calibration slope and net benefit. Clusters report stability (adjusted Rand index) and outcome rates with 95% intervals.

8 Five-week plan

WEEK	FOCUS	EXIT CRITERION
1	Environment, restructured repo, data download and cache, synthetic generator, cohort with fixed horizon	Tests pass on synthetic data; cohort counts match the source
2	Supervised baselines with validation split, consistent class weighting, site A ↔ B validation	Model comparison with CIs on real data
3	Unsupervised module: embeddings, clustering with stability, anomaly baseline	Planted subgroups recovered on synthetic data; real-data clusters profiled
4	CNN with masks, optional autoencoder, unified report	Report covers both families from one run
5	Streamlit application, documentation, full runs, write-up	App runs from saved artifacts; results reproducible from config

9 References

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